

AWS Developer Certification

Jacopo Pela

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The certification cover over **40 AWS services**
Creating resources in the cloud cost money.

Creating an AWS account requires an email for the root user and an AWS account name. You also need a password to authenticate with your root email.

Cloud computing is the on-demand delivery of compute power, database storage, applications, and other IT resources.
Through a cloud services platform with pay-as-you-go pricing
You can provision exactly the right type and size of computing resources you need

AWS is one of the many cloud providers available on the market!

The five characteristics of cloud computing are the following:

- **On-demand services:** without the need of human interaction
- **Broad network access:** resources can be accessed everywhere
- **Multi-tenancy and resource pooling** multiple customers are served on the same resources
- **Rapid elasticity and scalability** easily create and dispose of resources
- **Measured service** users pay for what they use

AWS has 3 pricing fundamentals:

- **Compute:** Pay for your computational time
- **Storage:** Pay for data stored
- **Data transferred OUT of the cloud**

For the six advantages of Cloud Computing remember: **Trade capital expense for variable expense, Increase speed and agility**

Shared responsibility Model

You are responsible for the security **IN** the cloud

AWS is responsible for the security **OF** the cloud

AWS global services are: **IAM, CloudFront, WAF and Route53, Workspace**

Software as a Service (SaaS)

Software as a Service provides you with a completed product that is run and managed by the service provider. In most cases, people referring to Software as a Service are referring to end-user applications. With a SaaS offering, you do not have to think about how the service is maintained or how the underlying infrastructure is managed; you only need to think about how you will use that particular piece of software.

Platform as a Service (PaaS) Platform as a Service (PaaS) as a service removes the need for organizations to manage the underlying infrastructure (usually hardware and operating systems) and allow you to focus on the **deployment and management of your applications**. This helps you be more efficient as you don't need to worry about resource procurement, capacity planning, software maintenance , patching, or any of the other undifferentiated heavy lifting involved in running your application. Think PaaS like AWS, Azure and Openshift.

Available Zone (AZ) are **ONE** or more discrete datacenter in the same **region**

1 IAM

AWS IAM is one of the few **global** services

On AWS you apply the **least privilege** principle for the user and services IAM policy

1.1 IAM Security Tools

IAM Credentials Report (account-level) is a service used to lists of all your account's users and the status of their various credentials.

IAM Access Advisor (user-level) is a service used to shows the service permissions granted to a user and when those services were last accessed.

You can use this information to revise your policies.

Don't use the root account except for AWS account setup

Universal 2nd Factor (U2F) Security Key is a device that you can plug into a USB port on your computer.

Hardware Multi-Factor Authentication is a hardware device that generates a six-digit numeric code based upon a time-synchronized one-time password algorithm.

2 EC2

EC2 User Data script run with **root user**.

EC2 On-Demand instance are billed with the **pay per second** timeframe with a **one-minute minimum charge** based on the EC2 Instance

AWS has the following naming convention:

m5.2xlarge

- m: instance class
- 5: generation (AWS improves them over time)
- 2xlarge: size within the instance class

Usage beyond Savings Plans is billed at the On-Demand price.

An Amazon EC2 Dedicated Host allows you to use your eligible software licenses from vendors such as Microsoft and Oracle on Amazon EC2 so that you get the flexibility and cost-effectiveness of using your licenses, but with the resiliency, simplicity, and elasticity of AWS. An Amazon EC2 Dedicated Host is a physical server fully dedicated for your use, so you can help address corporate compliance requirements.

Dedicated Host support **Bring Your Own License (BYOL)** capability.

2.1 Security Groups

Security Groups control how traffic is allowed into or out of our instances.

Security groups only contain **allow** rules.

Security groups rules can reference by IP or by security group.

They controll both inbound and outbound traffic Security Group are linked to the specific **region and VPC** where they are created.

Security group roles are stateful, meaning they allow automatically the return traffic.

If your application is not accessible (time out), then it's a security group issue.

2.2 EBS Volume

They can only be mounted to one instance at a time.

They are bound to a specific availability zone.

It's locked to an Availability Zone, to move a volume you need to take a snapshot.

EBS volume can be attached to a single instance in the same Availability Zone (AZ). whereas EFS file system can be mounted on instances across multiple Availability Zones (AZ)

2.3 EBS Snapshots

Make a backup (snapshot) of your EBS volume at a point in time.

You can move your EBS snapshot to a cheaper archive tier.

You can specify a retention policy (1 day - 1 year) for your deleted EBS snapshot

EBS Snapshot archive are 75% cheaper

2.4 AMI

AMI are customization of an EC2 instance **EC2 Image Builder** is the service use to schedule and organize AMI creation

2.5 EC2 Image Builder

Used to automate the creation of Virtual Machines or container images.
Can be run on schedule.

Spot instances are available at up to a 90% discount compared to the on-demand instance prices

2.6 EC2 Instance store

high performance disk attached to EC2 instances

2.7 EFS

EFS can be mounthed to multiple target (100s of EC2 at the same time) and is not available zone bounded.

Amazon EC2 instances can access your file system across AZs, regions, and VPCs, while on-premises servers can access using AWS Direct Connect or AWS VPN.

EFS Infrequent Access (IA) is 92% cheaper compared to EFS Standard.
AWS EFS can be used with on-premise system.

2.8 AWS FSx

Launch 3th party high performance sys on AWS.
FSx is a windows native shared file system.

High availability means running your application / system in at least 2 Availability Zones

Elasticity is the ability to acquire resources as you need them and release resources when you no longer need them. In the cloud, you want to do this automatically.

Agility is the ability to rapidly develop, test and launch software applications that drive business growth.

3 AWS S3

Key-value based object storage

Maximum size of an s3 object is 50TB.

For uploading object bigger than 5Gb we need to use **multi-part upload**.

S3 bucket policy are used for cross-account bucket/object sharing.

Data transferred out is free when it's transferred to a AWS service in the same AZ

S3 bucket versioning is enabled at bucket level.

Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA) is for data that is accessed less frequently but requires rapid access when needed.

Amazon S3 Glacier Flexible Retrieval (formerly S3 Glacier) is the ideal storage class for archiving data that does not require immediate access but needs the flexibility to retrieve large sets of data at no cost, such as backup or disaster recovery use cases.

3.1 IAM Access Analyzer for S3

Ensure that only the correct people have access to your S3 bucket

3.2 AWS Snowball

On-edge device to **collect and process** data and **migrate** them on AWS.

3.3 AWS OpsHub

AWS OpsHub is a graphical user interface you can use to manage your AWS Snowball devices, enabling you to rapidly deploy edge computing workloads and simplify data migration to the cloud.

3.4 AWS Storage Gateway

Bridge between the on premise data and AWS data storage option (S3, EBS,...).

Hybrid storage service to allow on-premises to seamlessly use the AWS Cloud. AWS Storage Gateway service provides three different types of gateways – **Tape Gateway, File Gateway, and Volume Gateway** that seamlessly connect on-premises applications to cloud storage, caching data locally for low-latency access.

3.5 S3 Express One Zone

AWS S3 storage fitted for operation that need millisecond retrieval time for stored objects.

Encryption in transit and at rest are a client responsibility for AWS S3

S3 Intelligent tier **don't charge** for data retrieval

4 Database solutions

4.1 AWS RDS

RDS is a **Relational** Database that use SQL data formatting.

Amazon Aurora RDS engine have the **serverless** option, where we don't provision a capacity for the database but the size is automatically managed by AWS.

When you provision an Amazon RDS Multi-AZ Instance with one standby, Amazon RDS automatically creates a primary DB Instance and synchronously replicates the data to a standby instance in a different Availability Zone (AZ).

Read replica need to be **manually approved** to be its own standalone DB instance.

Read Replica improves **database scalability**.

4.2 ElastiCache

ElastiCache is the managed cache solution for AWS.

AWS takes care of OS maintenance / patching, optimizations, setup, configuration, monitoring, failure recovery and backups

4.3 DynamoDB

The noSQL database service for AWS with serverless database capability.

Fully Managed Highly available with replication across 3 AZ.

Integrated with IAM for security, authorization and administration.

Standard and IA table class.

DAX is the in-memory cache service for dynamoDB used only for the database service.

You can use Amazon DynamoDB to store recommendation results with the **LEAST operational overhead** for any scale.

4.4 Global Tables

Make a DynamoDB database accessible with low latency on multiple region.

DynamoDB global tables replicate data automatically across your choice of AWS Regions and automatically scale capacity to accommodate your workloads.

4.5 Redshift

Amazon Redshift is a fully-managed petabyte-scale cloud-based data warehouse product designed for large scale data set storage and analysis.

OLAP (online analytical process) database with serverless capability.

Redshift can be deployed with serverless capability

4.6 Amazon EMR

Elastic MapReduce DB service that manage hundred of EC2 instances with the possibility to fire "spot instances" used for processing large batch of data streamlined from ML and big data operations.

Use cases: data processing, machine learning, web indexing, big data...

4.7 Amazon Athena

Serverless database service used to analyze big batch of data with SQL query from object stored on S3

4.8 Amazon QuickSight

Serverless machine learning-powered service use to create interactive dashboard

4.9 DocumentDB

MongoDB inspired service used to store document.

Fully Managed, highly available with replication across 3 AZ with storage that automatically grows in increments of 10GB

4.10 Amazon Neptune

Fully managed **graph** database Highly available across 3 AZ with up to 15 read replicas.

4.11 Amazon Glue

Fully serverless and managed ETL service.

4.12 Amazon Timestream

Fully managed serverless **time series** database

4.13 Amazon Lightsail

Virtual servers, storage, databases, and networking.

Good for people with little cloud experience Amazon Lightsail is an easy-to-use cloud platform that offers you everything needed to build an application or website, plus a cost-effective, monthly plan. Amazon Lightsail offers several preconfigured, one-click-to-launch operating systems, development stacks, and web applications.

Has high availability but no auto-scaling, limited AWS integrations

Use case: Dev/Test env, Website and Simple web app

5 Infrastructure at scale

5.1 AWS System Manager - SSM

AWS Systems Manager gives you visibility and control of your infrastructure on AWS. Systems Manager provides a unified user interface so you can view operational data from multiple AWS services and allows you to automate operational tasks such as collecting software inventory, running commands, managing patches, and configuring servers across AWS Cloud as well as on-premises infrastructure.

SSM Service help manage the EC2 instance and on-premise server.

6 AWS Budget

You can use AWS Budgets to set reservation utilization or coverage targets and receive alerts when your utilization drops below the threshold you define. Under AWS Budget you can create reservation for: **Cost Budget, Reservation Budget and Usage Budget**

7 AWS Trusted Advisor

AWS Trusted Advisor is an online tool that provides you real-time guidance to help you provision your resources following **AWS best practices** on:

- cost optimization
- security
- fault tolerance
- service limits
- performance improvement.

For the Business & Enterprise Support plan:

- Full Set of Checks
- Programmatic Access using AWS Support API

8 CloudTrail

By default, the log files delivered by CloudTrail to your S3 bucket are encrypted using server-side encryption with Amazon S3 managed keys (SSE-S3). CloudTrail events provide a history of both API and non-API account activity made through the **AWS Management Console**, AWS SDKs, command line tools, and other AWS services.

Think account-specific activity and audit; think CloudTrail.

Management Event are logged by default on CloudTrail, while data and Insight must be enabled.

9 AWS Support

You should use the **AWS Developer Support** plan if you are testing or doing early development on AWS and want the ability to get **email based technical support** during business hours as well as general architectural guidance as you build and test.

You should use **AWS Business Support** if you have production workloads on AWS and want 24x7 phone, email and chat access to technical support and architectural guidance in the context of your specific use-cases.

You should use the **AWS Enterprise On-Ramp Support** plan if you have production/business critical workloads.

The **AWS Abuse team** can assist you when AWS resources are used to engage in abusive behavior.

The **Acceptable Use Policy** describes prohibited uses of the web services offered by Amazon Web Services

9.1 APN Consulting Partner

The AWS Partner Network (APN) is the global partner program for technology and consulting businesses that leverage Amazon Web Services to build solutions and services for customers.

9.2 APN Technology Partner

APN Technology Partners provide hardware, connectivity services, or software solutions that are either hosted on or integrated with, the AWS Cloud. APN

10 AWS WAF

WAF protect against common exploits at **Layer 7**.

AWS WAF can be deployed on the following services: **Amazon CloudFront, Application Load Balancer, Amazon API Gateway, AWS AppSync**

10.1 AWS Network Firewall

Protect your entire Amazon VPC from Layer 3 to Layer 7 protection, for any direction you can inspect

11 Amazon Shield

AWS Shield Advanced provides expanded DDoS attack protection for web applications running on the following resources:

- Amazon EC2
- Elastic Load Balancing (ELB)
- Amazon CloudFront
- Amazon Route 53
- AWS Global Accelerator.

12 CloudFront

AWS Edge location is a site that CloudFront uses to cache copies of the content for faster delivery to users at any location.

AWS CloudFront is a **global service** in scope.

CloudFront Origins are: **S3 Bucket, VPC Origin, Custom Origin.**

12.1 S3 Transfer Acceleration

Increase transfer speed by transferring file to an AWS edge location which will forward the data to the S3 bucket in the target region

12.2 AWS WaveLength

WaveLength Zones are infrastructure deployments embedded within the telecommunications providers' datacenters at the edge of the 5G networks

13 ElasticBeanstalk

With basic health reporting, the AWS Elastic Beanstalk service **does not publish any metrics to Amazon CloudWatch**

14 VPC

ALB - HTTP/HTTPS traffic

NLB - TCP/UDP traffic

A **VPC Gateway Endpoint** is a gateway that you specify as a target for a route in your route table for traffic destined to a supported AWS service. The following AWS services are supported:

- Amazon Simple Storage Service (Amazon S3)
- Amazon DynamoDB

You may see a question around this concept in the exam. Just remember that only Amazon S3 and Amazon DynamoDB support VPC gateway endpoint. All other services that support VPC Endpoints use a VPC interface endpoint. **VPC Endpoint Interface** are used for the remaining AWS services
NACL rule only include IP address.

14.1 Internet Gateway

An Internet Gateway is a horizontally scaled, redundant, and highly available VPC component that allows communication between instances in your VPC and the internet. Therefore, it imposes no availability risks or bandwidth constraints on your network traffic.

Internet Gateway communication **use the public internet.**

14.2 NAT Gateways

NAT Gateways (**AWS-managed**) and NAT Instances (**self-managed**) allow your instances in your Private Subnets to access the internet while remaining private.

NAT Gateway are managed by AWS

14.3 AWS Private Link

AWS PrivateLink enables private connectivity between VPCs and supported AWS services without traffic traversing the public internet. It ensures secure communication for applications and services.

Most secure & scalable way to expose a service to 1000s of VPCs.

Does not require VPC peering, internet gateway, NAT, route tables., but requires a network load balancer and ENI.

14.4 AWS Site-to-site

AWS Site-to-Site VPN creates a secure connection between your data center or branch office and your AWS cloud resources. **This connection goes over the public internet.**

Virtual private gateway (VGW)/Transit Gateway and Customer Gateway are the components of an AWS Site-to-Site VPN

14.5 Customer gateway

A customer gateway device is a physical or software appliance on your side of a Site-to-Site VPN connection. You or your network administrator must configure the device to work with the Site-to-Site VPN connection. You can enable access to your remote network from your VPC by creating an AWS Site-to-Site VPN (Site-to-Site VPN) connection, and configuring routing to pass traffic through the connection.

14.6 VPC Flow Logs

Capture information about IP traffic going into your interfaces.

Captures network information from AWS managed interfaces too: Elastic Load Balancers, ElastiCache, RDS, Aurora, etc. . .

14.7 VPC Peering

Connect two VPC, privately using AWS' network.

VPC Peering connection is **not transitive**

14.8 Direct Connect

AWS Direct Connect creates a dedicated private connection from a remote network to your VPC. This is a private connection and does not use the public internet. Takes at least a month to establish this connection. **AWS Direct Connect cannot be used to interconnect VPCs.**

14.9 Transit Gateway

Starhub service that connect thousands of VPC and on-premises networks together

14.10 AWS Local Zones

AWS Local Zones are a type of AWS infrastructure deployment that places AWS compute, storage, database, and other select services close to a large population, industry, and IT centers. With AWS Local Zones, you can easily run applications that need single-digit millisecond latency closer to end-users in a specific geography.

15 AWS Global Accelerator

AWS Global Accelerator is a service that improves the availability and performance of your applications with local or global users. It provides static IP addresses that act as a fixed entry point to your application endpoints in a single or multiple AWS Regions, such as your Application Load Balancers, Network Load Balancers, or Amazon EC2 instances. AWS Global Accelerator uses the AWS global network to optimize the path from your users to your applications, improving the performance of your traffic by as much as 60%.

16 Security

16.1 AWS Shield

Shield standard is a free service activated by default.

Shield Advance provide protection over DDOS attack 24/7 with a monthly fee of 3000\$ and more sophisticated attack.

AWS Shield protect: **EC2, ELB, Route53, CloudFront and Global Accelerator**

16.2 AWS Firewall Manager

Manage security rules in all accounts of an AWS Organization.

Rules are applied to new resources as they are created across all and future accounts in your Organization.

Encryption is enabled by default on **CloudTrail log, AWS S3 glacier and Storage Gateway.**

16.3 AWS Certificate Manager

ACM lets you provision, manage, and deploy **TLS/SSL** certificates for AWS service.

ACM is free of charge for public TLS certificates.

16.4 AWS Artifact

Portal that provides customers with on-demand access to AWS compliance documentation and AWS agreements.

16.5 AWS Architecture Center

The AWS Architecture Center provides reference architecture diagrams, vetted architecture solutions, Well-Architected best practices, patterns, icons, etc.

16.6 Amazon GuardDuty

Intelligent Threat discovery to protect your AWS Account that uses Machine Learning algorithms, anomaly detection, 3rd party data.

Can protect against Cryptocurrency attacks.

Amazon GuardDuty is a threat detection service that monitors malicious activity and unauthorized behavior to protect your AWS account. Amazon GuardDuty analyzes billions of events across your AWS accounts from AWS CloudTrail (AWS user and API activity in your accounts), Amazon VPC Flow Logs (network traffic data), and DNS Logs (name query patterns).

16.7 AWS Security Hub

Central security tool to manage security across several AWS accounts and automate security checks with an integrated dashboards showing current security and compliance status.

Automatically aggregates alerts in predefined or personal findings formats from various AWS services & AWS partner tools.

Must first enable the AWS Config Service.

Security Hub can aggregates information from service like Macie, GuardDuty, Inspector, ...

16.8 Amazon Detective

Amazon Detective analyzes, investigates, and quickly identifies the root cause of security issues or suspicious activities.

Amazon Detective can analyze events from multiple data sources such as: Virtual Private Cloud (VPC) Flow Logs, AWS CloudTrail, and Amazon GuardDuty

16.9 IAM Access Analyzer

Find out which resources are shared externally by defining Zone of Trust and notify when a resource is accessed outside a zone of trust

16.10 AWS Fault Injection Simulator (FIS)

A fully managed service for running fault injection experiments on AWS workloads based on Chaos Engineering that support EC2, ECS, EKS, RDS...

17 Machine Learning and AI

17.1 Amazon Transcribe

You can use Amazon Transcribe to add speech-to-text capability to your applications.

17.2 Amazon Polly

You can use Amazon Polly to turn text into lifelike speech thereby allowing you to create applications that talk.

17.3 Amazon CodeGuru

Amazon CodeGuru is a developer tool that provides intelligent recommendations to improve code quality and identify an application's most expensive lines of code.

17.4 Shared responsibility model

Under the AWS Shared Responsibility Model, customers are responsible for managing their data, **including data encryption**. AWS trains AWS employees, but a customer must train their own employees. AWS is responsible for patching and fixing flaws within the infrastructure, but customers are responsible for patching their guest operating system and applications.

17.5 Amazon Elastic Transcoder

Amazon Elastic Transcoder lets you convert media files that you have stored in Amazon Simple Storage Service (Amazon S3) into media files in the formats required by consumer playback devices. For example, you can convert large, high-quality digital media files into formats that users can playback on mobile devices, tablets, web browsers, and connected televisions.

17.6 Amazon Macie

Amazon Macie is a fully managed data security and data privacy service that uses machine learning and pattern matching to **discover and protect your sensitive data in AWS**. Macie automatically provides an inventory of Amazon S3 buckets including a list of unencrypted buckets, publicly accessible buckets, and buckets shared with AWS accounts outside those you have defined in AWS Organizations.

Amazon Macie automatically detects a large and growing list of sensitive data types, including personally identifiable information (PII)

17.7 **Amazon Kendra**

Fully managed document search service powered by Machine Learning that extract answers from within a document

17.8 **Amazon Personalize**

Fully managed ML-service to build apps with real-time personalized recommendations.

17.9 **AWS Resource Access Manager**

Share AWS resources that you own with other AWS accounts or with any within your Organization by avoiding resource duplication.

17.10 **Amazon Comprehend**

Amazon Comprehend is a natural-language processing (NLP) service that uses machine learning to uncover information in unstructured data. Instead of combing through documents, the process is simplified and unseen information is easier to understand.

17.11 **Amazon Pinpoint**

Scalable 2-way (outbound/inbound) marketing communications service that supports email, SMS, push, voice, and in-app messaging.

Use cases: run campaigns by sending marketing, bulk, transactional SMS messages

17.12 Amazon Forecast

Amazon Forecast is a fully managed service that uses machine learning to deliver highly accurate forecasts. Amazon Forecast can be used to forecast any time-series data, such as retail demand, manufacturing demand, travel demand, revenue, IT capacity, logistics, and web traffic.

18 Well-architected framework

The AWS Well-Architected Framework provides guidance on building secure, high-performing, resilient, and efficient infrastructure for cloud based applications. Based on six pillars:

- operational excellence
- security
- reliability
- performance efficiency
- cost optimization
- sustainability

18.1 Well-Architected tool

The AWS Well-Architected Tool helps you review the state of your workloads and compares them to the latest AWS architectural best practices. The tool is based on the AWS Well-Architected Framework, developed to help cloud architects application infrastructure. To use this free tool, available in the AWS Management Console, just define your workload and answer a set of questions.

The AWS Well-Architected Tool then provides a plan on how to architect for the cloud using established best practices.

18.2 AWS CAF

AWS CAF groups its capabilities in six perspectives: **Business, People, Governance, Platform, Security, and Operations**

The Business Perspective of the AWS CAF ensures that cloud adoption strategies align with the organization's business objectives. It focuses on strategic planning, value realization, and benefits tracking to ensure cloud adoption drives business success.

The Governance Perspective focuses on managing risks, compliance, and financial management in the cloud. While important, it does not directly address aligning IT strategy with business objectives.

The Operations Perspective emphasizes day-to-day workload operations, monitoring, and performance management. It does not focus on strategy alignment with business goals.

The Platform Perspective is centered on designing, implementing, and optimizing cloud services and applications. Helps you build an enterprise-grade, scalable, hybrid cloud platform; modernize existing workloads; and implement new cloud-native solutions.

Security Perspective helps you achieve the confidentiality, integrity, and availability of your data and cloud workloads.

Operations Perspective helps ensure that your cloud services are delivered at a level that meets the needs of your business

AWS Partner Solutions (formerly Quick Starts) AWS Partner Solutions are automated reference deployments for key workloads on the AWS Cloud. Each Partner Solution launches, configures, and runs the AWS compute, network, storage, and other services required to deploy a specific workload on AWS, using AWS best practices for security and availability.

19 Service Health

While the AWS Health - Service Health Dashboard displays the general status of AWS services; the AWS Health - Your Account Health Dashboard gives you a personalized view of the performance and availability of the AWS services underlying your AWS resources.

20 AWS Config

AWS Config is a service that enables you to assess, audit, and evaluate the configurations of your AWS resources. Config **continuously monitors and records** your AWS resource configurations and allows you to automate the evaluation of recorded configurations against desired configurations. Think resource-specific history, audit, and compliance; think Config.

21 Amazon Compute Optimizer

AWS Compute Optimizer helps you identify the optimal AWS resource configurations, such as:

- Amazon EC2 instance types
- Amazon EBS volume configurations
- AWS Lambda function memory sizes

22 Other services

22.1 AWS Control Tower

Easy way to set up and govern a secure and compliant multi-account AWS environment based on best practices.

AWS Control Tower is an AWS native service providing a pre-defined set of blueprints and guardrails to help customers implement a landing zone for new AWS accounts.

AWS Control Tower is designed to provide an easy, self-service setup experience and an interactive user interface for ongoing governance with guardrails.

While Control Tower automates creation of a new landing zone with pre-configured blueprints

22.2 AWS Infrastructure Composer

Visually design and build serverless applications quickly on AWS and allow to deploy AWS infrastructure code without needing to be an expert in AWS.

22.3 AWS Device Farm

Fully-managed service that tests your web and mobile apps against desktop browsers, real mobile devices, and tablets

22.4 Reliability

Ability of a system to recover from infrastructure or service disruptions, dynamically acquire computing resources to meet demand, and mitigate disruptions such as misconfigurations or transient network issues

22.5 AWS Managed Services

Provides infrastructure and application support on AWS. AMS offers a team of AWS experts who manage and operate your infrastructure for security, reliability, and availability.

22.6 AWS Cost & Usage Report (AWS CUR)

AWS Cost & Usage Report (AWS CUR) contains the most comprehensive set of cost and usage data available. You can receive reports that break down your costs by the hour or month, by product or product resource, or by tags that you define yourself.

AWS updates the report in your bucket once a day in comma-separated value (CSV) format.

CUR is used to publish your AWS billing reports to an Amazon Simple Storage Service (Amazon S3) bucket that you own

22.7 AWS Price Calculator

Power/Cooling are included in the server, storage, and network cost.

22.8 AWS Cost Explorer

AWS Cost Explorer has an easy-to-use interface that lets you visualize, understand, and manage your AWS costs and usage over time. AWS Cost Explorer includes a default report that helps you visualize the costs and usage associated with your top five cost-accruing AWS services, and gives you a detailed breakdown of all services in the table view.

AWS Cost Explorer also supports forecasting to get a better idea of what your costs and usage may look like in the future.

Every AWS account provides its own invoice end of the month. You can get separate invoices for development and production environments by setting up separate AWS accounts for each environment.

EC2 Savings Plan - Up to 72% discount compared to On-Demand
Compute Savings Plan - Up to 66% discount compared to On-Demand for EC2, fargate and lambda

23 CloudWatch

AWS CloudWatch is able to monitor the following metrics:

- Application performance
- Resource utilization
- System-wide visibility
- Unified operational health

24 AWS DataSync

AWS DataSync is a secure online data transfer service that simplifies, automates, and accelerates copying terabytes of data to and from AWS storage

services. Easily migrate or replicate large data sets without having to build custom solutions or oversee repetitive tasks.

You can use AWS DataSync for ongoing transfers from on-premises systems into or out of AWS for processing. DataSync can help speed up your critical hybrid cloud storage workflows in industries that need to move active files into AWS quickly.

24.1 AWS Backup

Centralized service for the backup routine of various AWS service with **Point-in-time** recovery capability.

With **Cross-Region** and **Cross-Account** backup capability

24.2 Migration Evaluator

Helps you build a data-driven business case for migration to AWS by providing a clear baseline of what your organization is running today.

Work by installing Agentless Collector to conduct broad-based discovery and analyze current state, define target state, then develop migration plan.

25 CloudEndure Disaster Recovery

CloudEndure Disaster Recovery, available from the AWS Marketplace, continuously replicates server-hosted applications and server-hosted databases from any source into AWS using block-level replication of the underlying server. CloudEndure Disaster Recovery enables you to use AWS Cloud as a disaster recovery Region for an on-premises workload and its environment. It can also be used for disaster recovery of AWS hosted workloads if they consist only of applications and databases hosted on EC2 (i.e. not RDS).

You may see use-cases asking you to select one of CloudWatch vs CloudTrail vs Config. Just remember this thumb rule:

Think resource performance monitoring, events, and alerts; think CloudWatch.

Think resource-specific change history, audit, and compliance; think Config.